RUSHI ASHOK SAVANI

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SUMMARY

AI/ML Engineer with 4+ years of experience building and deploying production-grade AI applications, agentic systems and machine learning solutions that solve complex business problems. Skilled in end-to-end software development, from identifying opportunities with stakeholders to architecting, optimizing and deploying reliable AI systems. A fast learner and hands-on problem solver who takes ownership of challenges, adapts quickly to new technologies, and focuses on delivering measurable business impact.

WORK EXPERIENCE

AI/ML Engineer | *ProHealth Group, Atlanta, Georgia* **Jun 2025 – Present**

- Collaborated with leadership and clinicians to identify **5+ AI automation opportunities** through operational analysis and analytical dashboards, proposing evidence-backed solutions for improving resource allocation and workflow inefficiencies.
- Architected and deployed a **Text-to-SQL agent** using **LangGraph, MCP** and hybrid retrieval, enabling natural-language queries across patient and operational records, reducing information retrieval time by **90%**.
- Automated **3+ operational and financial investigation workflows** using Python, FastAPIs, LLMs, and **LangChain**, delivering auditable root-cause reports that identified inefficiencies and saved **$90K** quarterly.
- Implemented **LLM evaluation and security frameworks** using LangSmith, RAGAS, MLflow and Postman to investigate the AI response against **100+ test cases** for accuracy, hallucinations, access control, and prevention of unauthorized database operations.
- Developed a patient **rehospitalization predictive model** on AWS **SageMaker** using Python, Scikit-learn and clinical feature engineering, distinguishing preventable high-risk cases to prioritize targeted intervention with **83%** recall.
- Validated model impact through causal inference and randomized **A/B testing** against standard care plans, cutting readmission cases by **10% (~$15K saving monthly)**.
- Deployed scalable AI services on AWS using **Kubernetes, docker and CI/CD**, reducing inference costs by **17%** through **Redis semantic caching**, incremental data processing and optimized LLM execution.

AI Research Assistant | *Syracuse University, New York* **Jan 2025 – May 2025**

- Analyzed **7,000+ employee survey responses** on enterprise AI adoption, automating data preprocessing to resolve missing values, standardize multilingual data, and prepare datasets for sentiment analysis.
- Fine-tuned **RoBERTa using PyTorch and Hugging Face Transformers** for study-specific sentiment classification, improving accuracy by **30%** while nearly eliminating manual labeling.
- Identified employee AI-adoption patterns using **Sentence-BERT embeddings** and **HDBSCAN clustering**, presenting behavioral segments and sentiment trends through **Tableau dashboards**.

Data Scientist | *IBM, India, Mumbai* **Jan 2022 - Jun 2024**

- Consulted **3 enterprise clients** on business challenges and ML opportunities, presenting **5+ solution proposals** decks to leadership and aligning recommendations with operational requirements.
- Delivered **risk-scoring and demand-forecasting models** using XGBoost, Random Forest, and Logistic Regression on Azure ML, improving baseline precision by **18%** and contributing to **$1.2M in operational savings**.
- Established end-to-end **ETL pipelines** using Azure Data Factory and SQL, integrating multiple data sources into staging and analytical marts to process **1M+ daily transactions** with **99.9% availability**.
- Assessed ML model performance through **real-world testing, statistical validation, and Bayesian optimization** using Optuna, addressing data leakage and accelerating model selection by **20%**.
- Communicated findings through **Tableau dashboards and executive presentations**, translating model outcomes into recommendations for non-technical stakeholders while improving internal PostgreSQL query performance by **40%**.

TECHNICAL SKILLS

- **Languages & Development:** Python, SQL, TypeScript, Node.js, FastAPI, Git Version control, Claude Code, Cursor, Codex
- **Generative AI & Agents:** LLMs, RAG, MCP, Agentic AI, LangChain, LangGraph, Prompt Engineering, LlamaIndex, Google ADK, Hybrid Retrieval, Tool calling
- **ML & Deep Learning:** Scikit-learn, PyTorch, Hugging Face, Pandas, NumPy, XGBoost, Regression, Classification, Hyperparameter Tuning, Classification Metrics, Natural Language Processing (NLP), Sentiment Analysis, Feature Engineering
- **Inference, Evaluation & MLOps:** vLLM, LiteLLM, CUDA, GPU optimization, LangSmith, RAGAS, MLflow, Model Validation, Guardrails, Docker, Kubernetes, GitHub Actions CI/CD, Redis, CloudWatch, Token-economics
- **Cloud & Data Engineering:** AWS (SageMaker, Bedrock, ECS, SQS, RDS, IAM), Microsoft Azure (Azure ML, Data Factory), SQL Server, PostgreSQL, Snowflake, Vector Store (pgvector), ETL pipelines, Data modeling
- **Analytics:** Power BI, Tableau, Streamlit, Statistical Analysis, Hypothesis Testing, A/B Testing, Causal Inference

EDUCATION

Syracuse University, New York – Masters | Information Systems, Data Science **Aug 2024 - May 2026**

Thadomal Shahani Engineering College, Mumbai, India - Bachelors | Information Technology **Jun 2020 - Jun 2024**

PROJECTS

Distributed LLM Inference Platform | *Python, vLLM, PyTorch, GCP, Kubernetes*

- Optimized a **PyTorch** CUDA engine on NVIDIA L4 (naive, KV cache, continuous batching, vLLM) at 214 vs 34 tok/s at concurrency 16, cut O(n^2) decode to O(n) (~5x), FP16/INT8/INT4 (980 to 489 MiB), GKE HPA 1 to 4 pods. | GitHub

PrivateRAG | *Python, Qdrant, Llama 3.1, TypeScript, LlamaIndex*

- Designed a fully local **RAG** system for private-document Q&A with Qdrant vector indexing, metadata filtering, and Llama 3.1 inference, featuring source attribution and chunk-level traceability. | GitHub

AI Email Response System | *Python, RAG, MCP, LLM-as-a-Judge*

- Implemented an automated RAG-based email response workflow with policy checks, **MCP**-based routing, LLM-as-a-Judge validation, and human handoff for high-risk cases, reducing manual investigation by **70%**. | GitHub

CERTIFICATIONS

AI Fluency: Foundation & Frameworks by Anthropic | AWS Machine Learning Associate